

# Aby Saji Pappachan

Technical Lead

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GitHub: <https://github.com/AbyAbyss>

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## Profile

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Technical Lead / Senior Backend Engineer with 7+ years of experience delivering scalable SaaS platforms, high-performance backend systems, and data pipelines. Proven ability to lead end-to-end system design, optimise large-scale data workloads, and deliver production-ready solutions in fast-paced environments.

Strong expertise in **Node.js, Golang, Python, NestJS, AWS, Airbyte, DuckDB, Airflow, GraphQL, and LLM-based systems**. Experienced in building distributed systems, improving performance at scale, and driving architectural decisions across multiple projects.

Currently seeking **Backend · AI / LLMs · Platform Engineering** roles in the UK. **Legally eligible to work in the UK.**

## Professional Experience

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### Technical Lead, Josys Digital Technologies India Private Limited

Aug 2022 to Present

#### Senior Software Engineer

August 2022 – March 2025

#### Technical Lead

April 2025 – Present

- Promoted to **Technical Lead** based on sustained ownership of complex systems, architectural leadership, and delivery impact across multiple teams.
- Acted as **technical owner** for backend, platform, data, and AI initiatives supporting a large SaaS ecosystem.
- Led system design, implementation, and rollout of multiple business-critical services.

#### Data, Reporting & Analytics

- Designed and implemented a **modern reporting and analytics platform** using **Airbyte, DuckDB, and Apache Airflow**, enabling efficient ingestion, transformation, and querying of large datasets.
- Built **ETL pipelines and scheduled workflows** to support reporting use cases with improved performance and reduced infrastructure cost.
- Worked on analytical query optimisation and data modelling for reporting at scale.

#### AI, LLM & Graph Systems

- Designed and delivered **AI-powered internal platforms** using **LLMs** for context-aware data retrieval and intelligent workflows.
- Implemented **Graph Database-backed solutions** to model complex relationships and enable semantic querying.
- Built **MCP (Model Context Protocol) clients and servers** to standardise communication between AI agents and backend services.
- Integrated LLM workflows into existing systems to enhance automation and decision-making.

#### Backend & Platform Engineering

- Introduced **NestJS** as the company-wide backend standard, improving maintainability and reducing onboarding time for new engineers.
- Designed and built a **Notification Platform** using REST APIs, AWS SQS, Email, and Slack integrations.
- Developed a **Performance Testing Framework** to simulate large-scale SaaS environments, significantly reducing testing cost and improving system reliability.
- Built a **multi-database GraphQL platform** with automatic schema generation and dynamic filter conversion.
- Created a **cross-browser extension backend** supporting both Chromium-based browsers and Firefox with dynamic configuration.
- Developed **proxy servers** to decouple frontend and backend dependencies, accelerating UI team development cycles.
- Frequently consulted for **technical research, architecture reviews, and proof-of-concept development**.

## Recognition

- Recognised as a **top performer for two consecutive years** for full ownership of complex projects and consistent early delivery.
- Known internally as a go-to engineer for solving **ambiguous, high-impact technical problems**.

### Member of Technical Staff (MTS-1), InstaSafe, Bengaluru, India

May 2019 to Aug 2022

- Focused on R&D to introduce new secure features into existing enterprise products.
- Led a team of **3 engineers** delivering multiple security-focused enhancements.
- Designed and developed a **Zero Trust Network Access (ZTNA)** solution.
- Built a **cross-platform desktop application** using ElectronJS for secure access to private resources.
- Developed a **gateway server** enabling agentless access to private web applications.
- Implemented secure CRUD APIs and automated workflows using **Node.js and Spring Boot**.

### Software Engineer, GrowthEnabler, Bengaluru, India

Aug 2018 to May 2019

- Worked as a full-stack engineer on frontend and backend systems.
- Built an **SRM workflow platform** and a **Solution Source system** to manage startup-corporate collaborations.

### Developer, Haappy, Bengaluru, India

Jan 2018 to Jul 2018

- Developed multiple WebTV websites and a **CRM system** for task and workflow tracking.

## Education

### East Point College of Engineering and Technology, Bengaluru, India

Bachelor of Engineering (BE) - Computer Science, Jan 2013 to Jan 2017

## Technical Proficiencies

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|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Backend &amp; APIs:</b>           | Node.js, Node.js, TypeScript, TypeScript, NestJS, NestJS, Python (FastAPI), Python (FastAPI), Golang, Golang, REST APIs, REST APIs, GraphQL, GraphQL                                                 |
| <b>Cloud &amp; Infrastructure:</b>   | AWS (ECS, SQS, S3), AWS (ECS, SQS, S3), Docker, Docker, Cloud-native architectures, Cloud-native architectures                                                                                       |
| <b>Data &amp; Analytics:</b>         | Airbyte, Airbyte, DuckDB, DuckDB, Apache Airflow, Apache Airflow, ETL pipelines, ETL pipelines, Analytical data modelling, Analytical data modelling                                                 |
| <b>AI &amp; Intelligent Systems:</b> | Machine Learning, Machine Learning, LLMs, LLMs, LangChain, LangChain, Graph Databases, Graph Databases, MCP (Model Context Protocol), MCP (Model Context Protocol), Semantic search, Semantic search |
| <b>Databases:</b>                    | MySQL, MySQL, MongoDB, MongoDB, PostgreSQL, PostgreSQL, Graph Databases, Graph Databases                                                                                                             |
| <b>Frontend (Second ary):</b>        | React, React, Angular, Angular                                                                                                                                                                       |
| <b>Engineering Practices:</b>        | Distributed systems, Distributed systems, System design, System design, Performance testing, Performance testing, Backend architecture, Backend architecture                                         |